



SanMitra

Tech Solutions

SanMitra Synthetic Data Vault (SSDV) is a **journal-first** tool. It either **generates** internally consistent Indian trading-company books, or **reads** journals exported from another application, then shows **CEO / CFO / Board KPIs**, charts, and AI notes.

Client install (non-technical): [CLIENT_INSTALL.md](#) · Client manual (FAQs, support): [CLIENT_MANUAL.md](#) · Support: contact@sanmitratech.in · sanmitratech.in

GitHub (private): <https://github.com/jayamurli1954/SSDV>

1. What SSDV is and is not

SSDV is

- A double-entry vault. Every voucher is posted through ``post()``. Debit must equal credit or the voucher is rejected.
- A **read-only connector**. It copies journals **into** SSDV. It never writes back to Tally, Zoho, Busy, or MitraBooks.
- An OfficeMitra MIS screen (Streamlit) plus a PowerShell CLI.

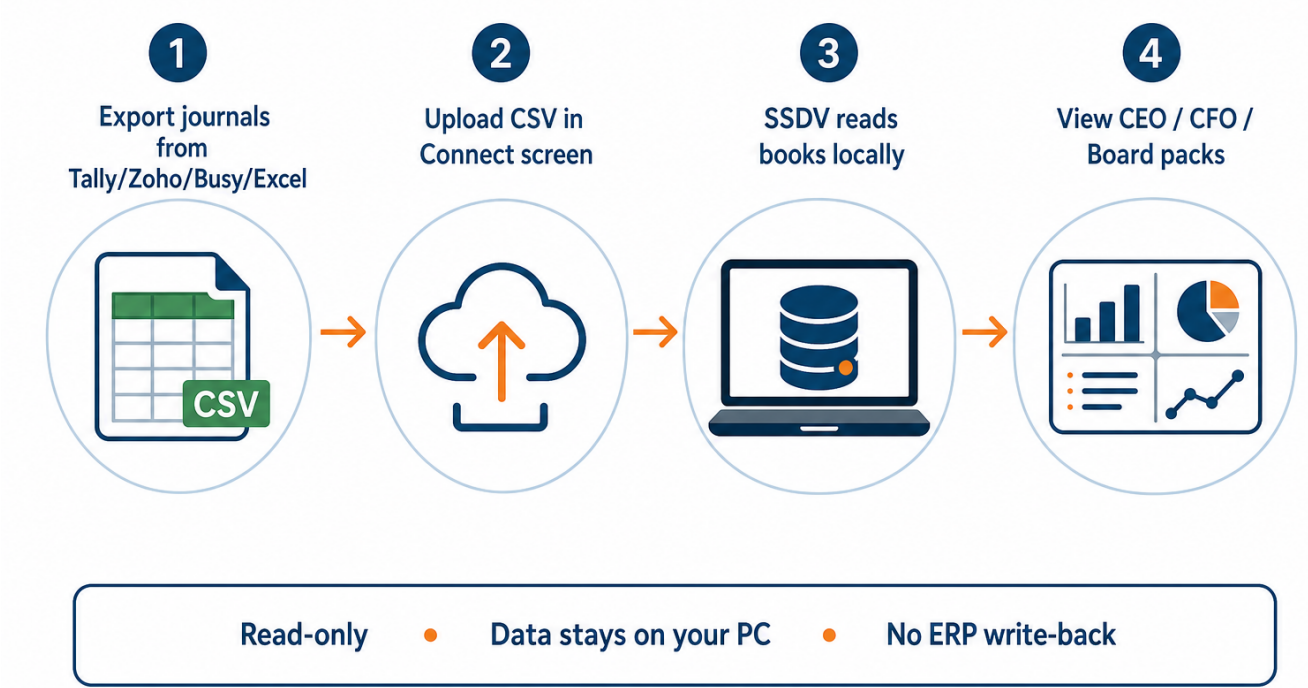
SSDV is not

- MitraBooks ERP, Tally, or a live login to any accounting product.
- A replacement for statutory books of record.
- Something you install into **global** Python (that broke ``sanmitra_unified-Next`` preflight once, because of a pytest version clash).

Two kinds of data:

Fuel	What it is	Typical vault	Typical as-of date
Generated books	ABC Industrial Supplies Pvt Ltd, FY 2023-24 to 2025-26	<code>`data\ssdv.sqlite`</code>	**2026-03-31**
Connected books	Your CSV day book / journal register	<code>`data\ssdv_connect.sqlite`</code>	Last date in that CSV (sample: **2024-04-30**)

OfficeMitra — How it works



2. Install (once, PowerShell)


SanMitra
TRUST · COMPLIANCE · GROWTH

Install OfficeMitra on your PC

- ✓

1

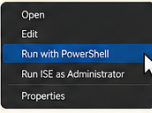
Unzip SSDV folder
 Extract the downloaded SSDV folder to any location on your PC.
 
- ✓

2

Install Python 3.11 if needed
 If Python 3.11 is not installed, download and install it from python.org


Python 3.11
- ✓

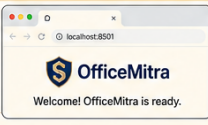
3

Right-click install.ps1
Run with PowerShell
 Right-click the file install.ps1 and select "Run with PowerShell".
 

- ✓

4

Double-click Start OfficeMitra
 Double-click the Start OfficeMitra file to launch the application.
 
- ✓

5

Browser opens localhost 8501
 Your default browser will open and OfficeMitra will be ready to use.
 


SanMitra
TRUST · COMPLIANCE · GROWTH

Smarter Tools. Stronger Compliance.
 Better Growth.

Use **only** the project virtual environment.

```
cd D:\SSDV
python -m venv .venv
.\.venv\Scripts\python.exe -m pip install -U pip
.\.venv\Scripts\python.exe -m pip install -e ".[dev]"
.\.venv\Scripts\python.exe -m pip install -e ".[ui]"
```

Do **not** run `pip install -e ".[ui]"` against `C:\Users\...\Python311\python.exe`.

2.1 Always call the venv `ssdv`

On this PC, typing `ssdv` often runs **global** Python, which has **no Streamlit**.

Use this instead, every time:

```
cd D:\SSDV
.\.venv\Scripts\ssdv.exe --help
```

Optional: activate the venv so `ssdv` in that window is the venv one:

```
cd D:\SSDV
.\.venv\Scripts\Activate.ps1
ssdv --help
```

If Activate is blocked, keep using `.\.venv\Scripts\ssdv.exe`.

2.2 Do not paste command *output* back into PowerShell

After `ssdv connect`, the tool prints lines like `Vouchers 7, Next: ssdv ...`. Those are messages, not commands. Only type lines you intend to run.

3. Fast start — ABC Industrial on screen

Generate (only if you do not already have `data\ssdv.sqlite`; **do not add `--force` unless you want to wipe the vault**):

```
cd D:\SSDV
.\.venv\Scripts\ssdv.exe init
.\.venv\Scripts\ssdv.exe validate --as-of 2026-03-31
```

Open the browser dashboard:

```
.\.venv\Scripts\ssdv.exe --db data/ssdv.sqlite ui --as-of 2026-03-31
```

Leave that window open. Open <http://localhost:8501> if the browser does not appear.

In the sidebar:

1. **Vault** → `D:\SSDV\data\ssdv.sqlite` (not `ssdv_connect.sqlite`).
2. Click **Use ABC year-end 31 Mar 2026**.
3. **Screen** → **CEO** (then CFO, Board, Charts as needed).

Stop the UI with **Ctrl+C** in that PowerShell window.

4. Fast start — sample connected books

This does **not** touch ABC Industrial.

```
cd D:\SSDV
.\.venv\Scripts\ssdv.exe connect --source generic --journals examples/generic_ingest/journals.csv --map exa
.\.venv\Scripts\ssdv.exe --db data/ssdv_connect.sqlite ui --as-of 2024-04-30
```

Sidebar: vault `ssdv_connect.sqlite`, as of **2024-04-30**, screen **CEO**.

5. Streamlit screens

OfficeMitra screens



Screen	What you see
CEO	14 KPI tiles, **top overdue customers** , benchmarks vs SSDV policy, **what-if scenarios (recommend-only)** , sales vs COGS, gross margin, collections, P&L mix, monthly profit, why-notes, why chips, optional typed question
CFO	Cash forecast 30/60/90 from open AR/AP (no sales plan), GST input/output/net, bank position, aging 0-30 through 120+, **top overdue customers and top vendor exposure** , DSO / DIO / DPO / CCC
Board	Red flags first, then assets / liabilities / equity, working-capital parts, policy scorecard, **benchmarks vs SSDV policy and ABC trading baseline** (not an industry survey), cash-cycle days, **top overdue / vendor tables**
Charts	Activity, sales vs COGS, cash movement, margin, P&L mix, monthly profit, aging
Connect	Upload a journal CSV + optional ledger map (read-only extract)

Download Board pack (PDF) writes an A4 pack (red flags, Board tiles, scorecard, benchmarks, 30/60/90 cash, what-if, top overdue customers, top vendor exposure). Journals are not changed. PPT is not in this pack. **Download full report (HTML)** is the CEO screen; open it in Edge and **Ctrl+P** if you want a print of that page.

All numbers come from **posted journals**. Unbalanced books never get a chart.

5.1 CEO Pack Dashboard

The **CEO Pack** provides executive visibility into overall business health, revenue momentum, liquidity, and operational levers:

Executive KPI Tiles & Performance Ribbon

14 primary KPI scorecards tracking revenue, gross profit, cash position, working capital, inventory, and collection efficiency.

OFFICEMITRA

Books

Vault

data/ssdv.sqlite

As of

2026/03/31

Use ABC year-end 31 Mar 2026

Screen

CEO pack

CFO pack

Board pack

Chart pack

Change data source

Change data source

Unbalanced books never get a chart. Same facts as ssdv mis --json.

> Full-page screenshot

OfficeMitra

Your AI CFO Dashboard

Read-only KPIs and AI notes from posted journals. Not a live login to Tally, Zoho, Busy, or MitraBooks.

POSTED BOOKS

ABC Industrial Supplies Private Limited

Scenario baselineAs of 2026-03-31Working capital 8,13,50,342.89

Download full report (HTML)

Download Board pack (PDF)

CEO PACK

Revenue increased 32.2% vs prior YTD.

Gross margin 20.5%. YTD profit (sales - COGS - operating expenses) -3,112,728.93.

Mar-26 profit (sales - COGS - operating expenses) -446,884.17.

Collection efficiency (receipts / sales) 102.0%. DSO 156.

AR 90+ is 60.4% of receivables (33,268,908.23 of 55,045,928.41).

Customer & Vendor Intelligence (Top Overdue Debtor Ranking)

Granular breakdown of customer receivables ranked by aging risk (AR 90+), unaged balances, and payment delays.

Metro Industrial Supplies has 406,657.13 in AR 90+ (0.9% of AR).

Largest vendor exposure is Precision Insulation Co 4,318,228.68 (6.4% of AP).

Operating working capital (AR + stock - AP) 81,350,342.89.

DSO is 156 days against a 120-day policy.

Cash in 90 days -79,835,831.05 if 0-90 day AR and AP convert. Unscheduled AR 33,268,908.23 is not timed.

SSDV policy breach: DSO, Current ratio, Quick ratio. Peer is ABC trading baseline when that vault exists, not an industry average.

FY2025-26 taxable sales

128,832,774.40

Gross profit %

20.5%

Cash position

-2,497,907.63

Receivables outstanding

55,045,928.41

Payables outstanding

66,911,608.22

Inventory value

93,216,022.70

Operating working capital

81,350,342.89

Mar-26 profit

-446,884.17

Collection efficiency

102.0%

Sales growth %

+32.2%

Cash conversion cycle

304 d

Current ratio

0.97x

SSDV Policy Benchmarks

Compares actual company performance against governance policy thresholds (DSO under 120 days, current ratio, quick ratio, stock holding).

Quick ratio
0.33x

Last-month net cash
-975,214.22

Customer & vendor intelligence

Invoice aging 90+ where invoices exist. Journal parties with no invoice date are unaged. Customers ranked by AR 90+ (or largest AR if none). Vendors ranked by AP outstanding. Recommend-only; SSDV does not chase customers or pay vendors.

Top overdue customers

Customer	AR 90+	Outstanding	Unaged	Oldest days
Metro Industrial Supplies	406657.13	479701.90	0.00	1088
Vertex Switchgear Traders Pune	367007.82	656916.47	0.00	1095
Pioneer Contractors Nagpur	364887.52	436471.19	0.00	978
Summit Switchgear Traders Kolhapur	354670.19	439010.73	0.00	979
Summit Power Solutions Solapur	310369.83	375098.48	0.00	1095
Kaveri Electricals Pvt Ltd Thane	310304.33	455760.50	0.00	1090
Summit Lighting House Navi Mumbai	291167.31	360710.80	0.00	815
Ujjwal Contractors Solapur	286670.78	425928.81	0.00	1040
Sahyadri Electricals Pvt Ltd Kolhapur	244744.43	244744.43	0.00	575
Vertex Lighting House Navi Mumbai	242902.59	286727.01	0.00	1095

What-If Scenarios & Recommendation Levers

Interactive simulation cards demonstrating potential cash and profit gains from operational improvements (e.g. reducing DSO to 120 days, 15% sales increase).

Benchmarks

SSDV policy bands for these trading books. Peer is ABC Industrial generated baseline (data/ssdv.sqlite) when that vault exists. Not a surveyed industry average.

KPI	Actual	SSDV policy	Policy	ABC baseline	vs ABC
Gross margin	20.5%	at least 16.0%	Hold	n/a	No ABC baseline vault
COGS / sales	0.80	under 0.84	Hold	n/a	No ABC baseline vault
DSO	156 d	under 120 days	Breach	n/a	No ABC baseline vault
Collection efficiency	102.0%	at least 80.0%	Hold	n/a	No ABC baseline vault
Current ratio	0.97x	at least 1.00x	Breach	n/a	No ABC baseline vault
Quick ratio	0.33x	at least 1.00x	Breach	n/a	No ABC baseline vault
Inventory / sales	0.31	under 0.45	Hold	n/a	No ABC baseline vault
Top customer share	1.0%	under 35.0%	Hold	n/a	No ABC baseline vault
Cash conversion cycle	304 d	no SSDV day cap	n/a	n/a	No ABC baseline vault

What-if (recommend-only)

Recommend-only. Each scenario scales posted YTD figures or open AR buckets. Books are not changed. No future sales plan beyond the stated lever.

If DSO improves to 120 days, how much cash is released?

If YTD sales rise 15%, what is projected profit?

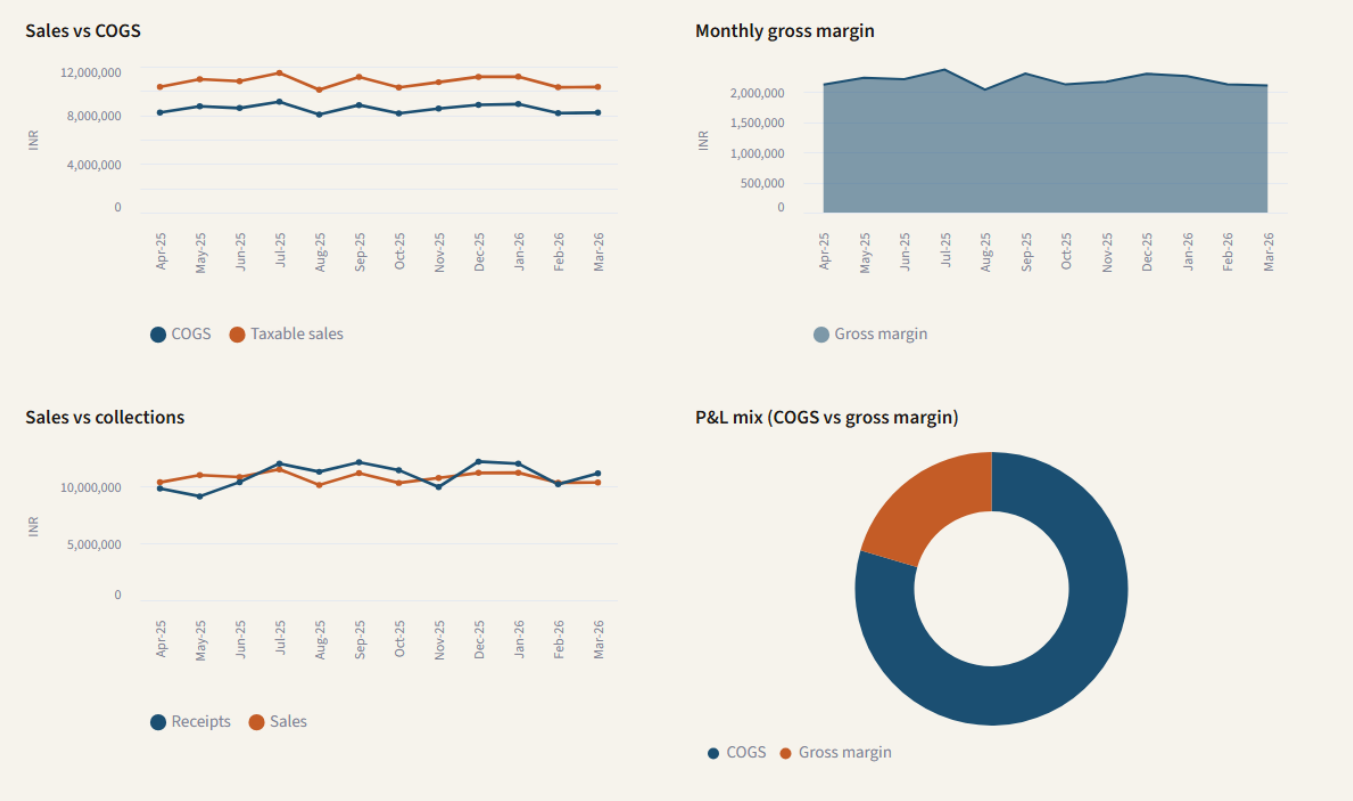
If AR 90+ is collected, how much cash is released?

If collection efficiency reaches 80%, how much extra cash?

Scenario	Result	Delta INR
If DSO improves to 120 days, how much cash is released?	DSO 156 days vs 120-day policy implies about 12,702,906.56 of AR (55,045,928.41 outstand	12702906.56
If YTD sales rise 15%, what is projected profit?	At the same gross margin and YTD opex (29,478,976.48), YTD profit moves from -3,112,728.	3954937.13
If AR 90+ is collected, how much cash is released?	Collecting open AR 90+ (33,268,908.23 of 55,045,928.41 receivables) releases that cash.	33268908.23
If collection efficiency reaches 80%, how much extra cash?	YTD receipts 131,394,168.09 are already at or above 80% of sales (128,832,774.40).	0.00

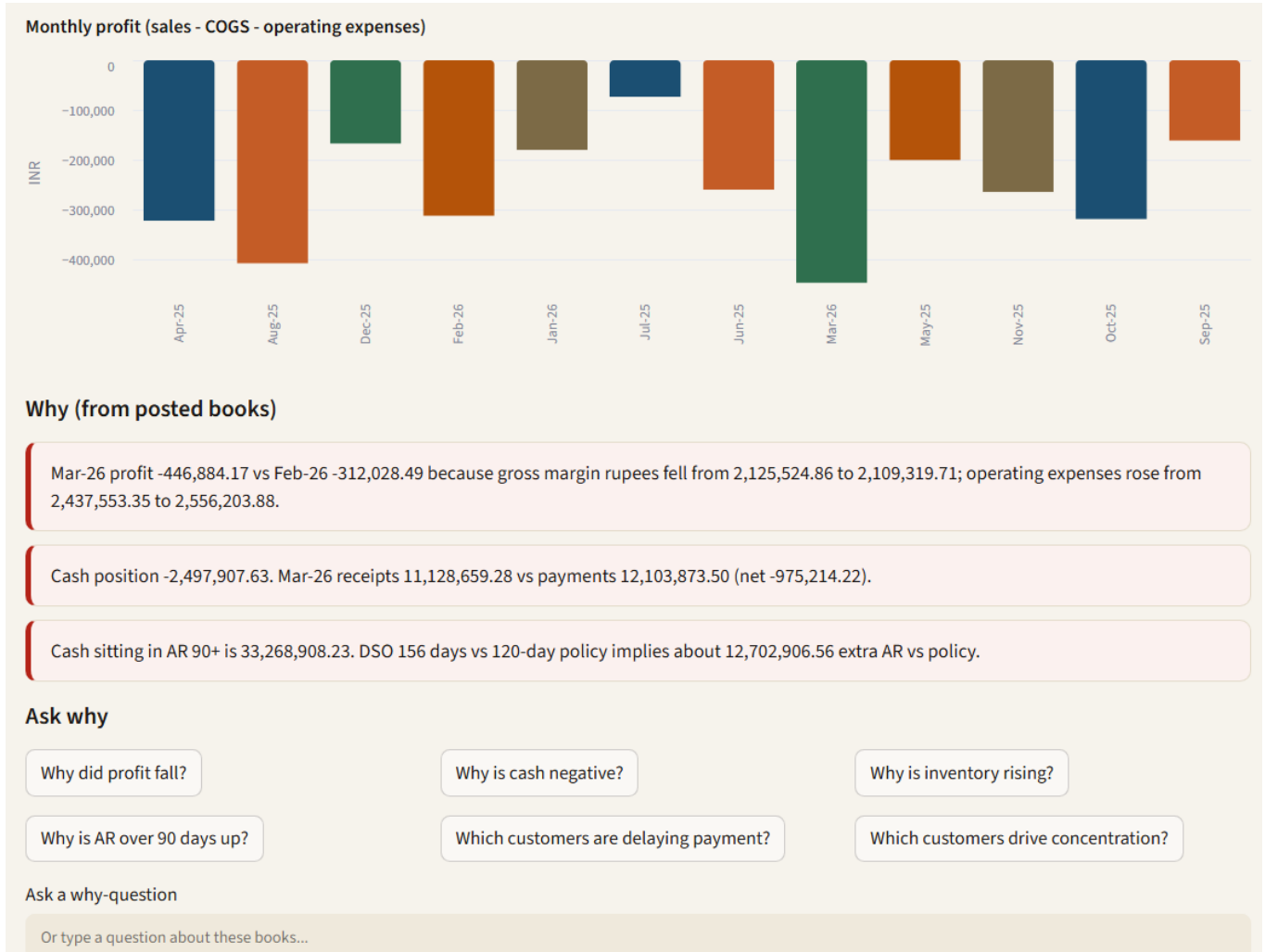
Sales, Margin & Cash Movement Charts

Trend charts showing monthly sales vs. COGS, gross margin stability, receipts vs. vendor payments, and P&L cost mix.



Plain-English AI Why-Notes & Copilot Queries

Instant explanations of profit movements, margin changes, and cash drains with interactive "Ask a Why Question" chips.



5.2 CFO Pack Dashboard

The **CFO Pack** is designed for deep financial control, liquidity management, and creditor reconciliation:

Working Capital & Liquidity KPI Tiles

Detailed balance sheet metrics including Debt/Equity ratio, Trade Receivables, Trade Payables, MSME 43B(h) overdue liabilities, and GST Net balances.

OFFICEMITRA

Books

Vault

data:ssdv.sqlite

As of

2026/03/31

Use ABC year-end 31 Mar 2026

Screen

☐ CEO pack

☒ CFO pack

☐ Board pack

☐ Chart pack

☐ Change data source

Change data source

Unbalanced books never get a chart. Same facts as `ssdv mis --json`.

OfficeMitra

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CFO PACK

DSO (AR / FY sales)

156 d

DuPont ROE

32.8%

DIO (stock / FY COGS)

332 d

DPO (AP / FY purchases)

184 d

Cash conversion cycle

304 d

Trade receivables

55,045,928.41

Trade payables

66,911,608.22

43B(h) MSME overdue risk

6,036,083.11

Inventory (WAC)

93,216,022.70

GST net liability

-3,216,846.57

Input GST

6,666,445.79

Output GST

3,449,599.22

Predictive 30 / 60 / 90-Day Cash Forecast

Forward-looking cash projection calculated strictly from open receivables (AR) and vendor payables (AP) timing buckets.

OD utilisation

100.0%

Salary payable

75,898,152.00

Cash in 30 days

-77,783,313.54

Cash in 60 days

-79,256,296.51

Cash in 90 days

-79,835,831.05

Cash forecast (open AR/AP, no sales plan)

Open AR and AP in 0-30 / 31-60 / 61-90 convert on those horizons. GST net liability and salary payable are 30-day outflows. AR 90+ and AR with no invoice aging are not timed. No future sales.

Cash today vs 30 / 60 / 90 days

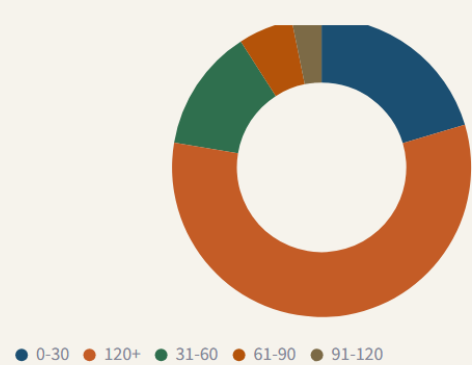
AR & AP Aging Distribution (Donuts & Buckets)

Visual aging analysis across 0-30, 31-60, 61-90, 90-120, and 120+ days for both debtors and creditors.

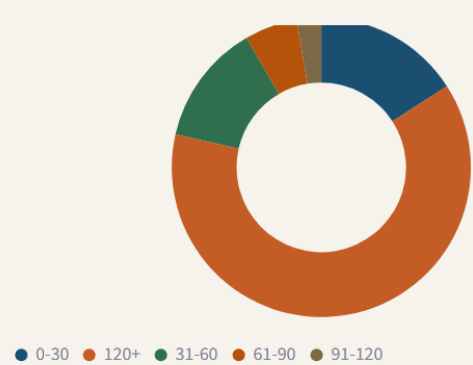
Horizon	AR in	AP out	GST out	Salary out	Net	Cash
30 days	11230984.02	10618237.93	0.00	75898152.00	-75285405.91	-77783313.54
60 days	7300574.84	8773557.81	0.00	0.00	-1472982.97	-79256296.51
90 days	3245461.32	3824995.86	0.00	0.00	-579534.54	-79835831.05

Unscheduled AR 3,32,68,908.23 is not in the 30/60/90 timing.

AR aging



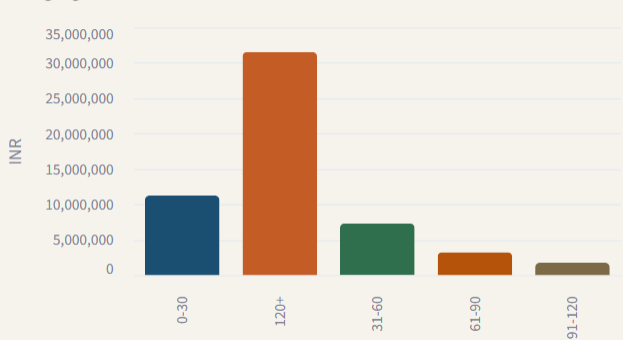
AP aging



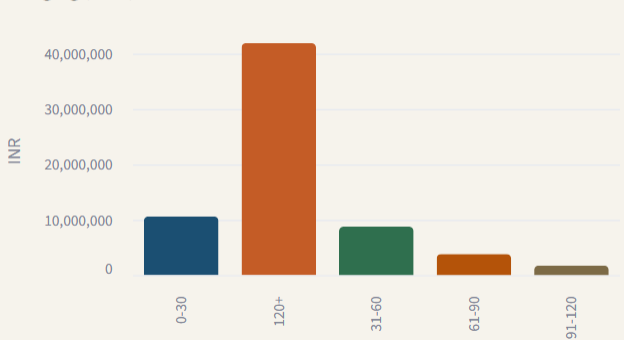
Customer & Vendor Intelligence

Simultaneous visibility into top overdue customer accounts and largest vendor exposures.

AR aging (bars)



AP aging (bars)



Cash Flow Receipts vs. Vendor Payments & Bank Position

Monthly operational receipts compared against supplier disbursements, alongside real-time bank current and overdraft (OD) positions.

Customer & vendor intelligence

Invoice aging 90+ where invoices exist. Journal parties with no invoice date are unaged. Customers ranked by AR 90+ (or largest AR if none). Vendors ranked by AP outstanding. Recommend-only; SSDV does not chase customers or pay vendors.

Top overdue customers

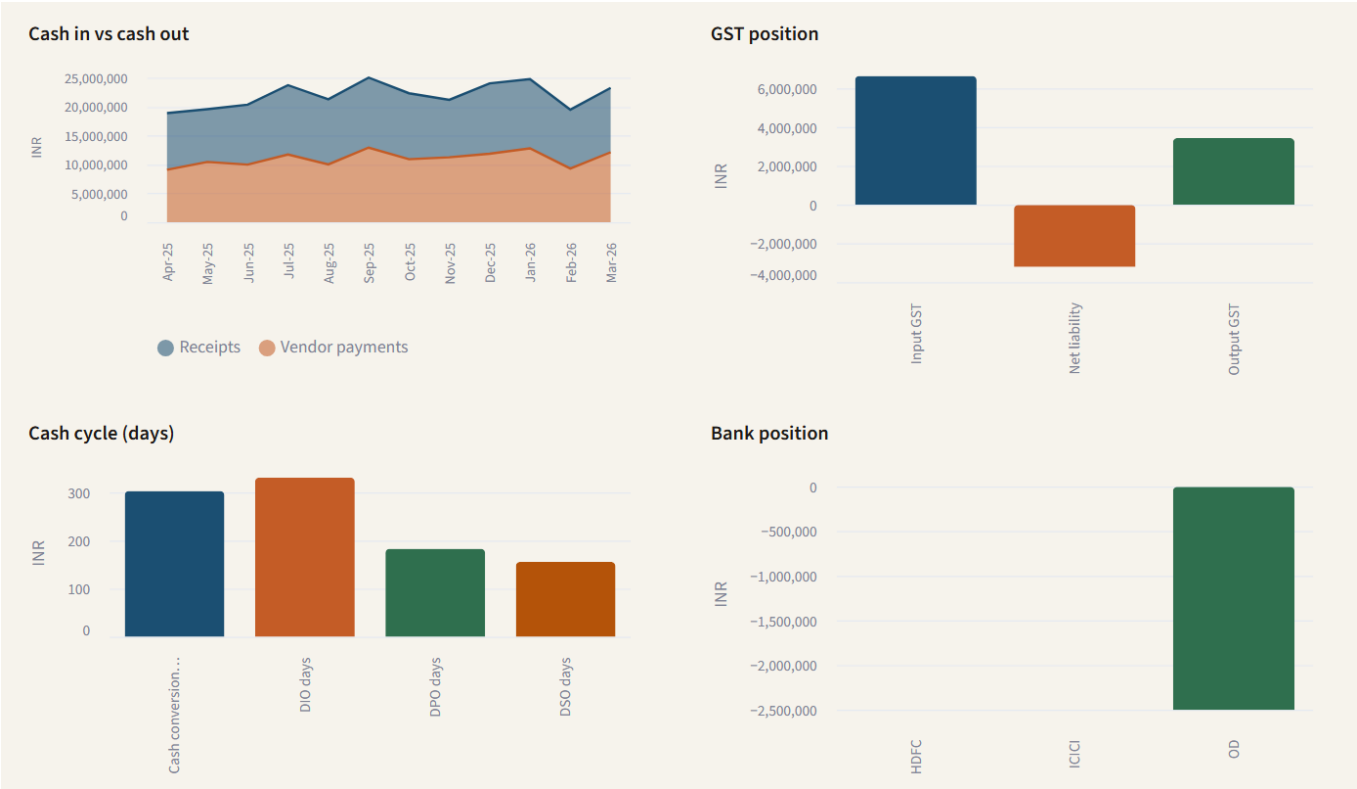
Customer	AR 90+	Outstanding	Unaged	O
Metro Industrial Supplies	406657.13	479701.90	0.00	
Vertex Switchgear Traders Pune	367007.82	656916.47	0.00	
Pioneer Contractors Nagpur	364887.52	436471.19	0.00	
Summit Switchgear Traders Kolhapur	354670.19	439010.73	0.00	
Summit Power Solutions Solapur	310369.83	375098.48	0.00	
Kaveri Electricals Pvt Ltd Thane	310304.33	455760.50	0.00	
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Ujjwal Contractors Solapur	286670.78	425928.81	0.00	
Sahyadri Electricals Pvt Ltd Kolhapur	244744.43	244744.43	0.00	
Vertex Lighting House Navi Mumbai	242902.59	286727.01	0.00	

Top vendor exposure

Vendor	Outstanding	AP 90+	Unaged	O
Precision Insulation Co	4318228.68	2594945.42	0.00	
Apex Polymers Pvt Ltd Kolhapur	4002062.15	2910379.18	0.00	
Continental Engineering Ltd	1713305.13	1103878.68	0.00	
Eastern Cable Mills Nashik	1682702.30	859797.43	0.00	
Universal Insulation Co Mumbai	1598378.70	1480308.42	0.00	
National Copper Products	1301704.99	750662.58	0.00	
Sterling Lamp Components	1277664.62	640419.94	0.00	
Reliable Polymers Pvt Ltd Nashik	1186063.06	678784.47	0.00	
Eastern Motor Industries Sangli	1184968.99	1041522.96	0.00	
Bharat Lamp Components Kolhapur	1117156.12	726183.66	0.00	

Cash Cycle Days & Statutory GST Balances

Granular breakdown of operating cash cycle days (DSO, DIO, DPO, CCC) and Input vs. Output GST net liabilities.



Automated Telemetry Alerts

Real-time red flags for collection efficiency dips, overdue debt concentration, and payment term violations.

Collection efficiency (receipts / sales) 102.0%. DSO 156.

AR 90+ is 60.4% of receivables (33,268,908.23 of 55,045,928.41).

Metro Industrial Supplies has 406,657.13 in AR 90+ (0.9% of AR).

Largest vendor exposure is Precision Insulation Co 4,318,228.68 (6.4% of AP).

DSO is 156 days against a 120-day policy.

Cash in 90 days -79,835,831.05 if 0-90 day AR and AP convert. Unscheduled AR 33,268,908.23 is not timed.

5.3 Board Pack Dashboard

The **Board Pack** delivers governance-ready telemetry and risk disclosures formatted for directors and board meetings:

Governance Red Flags & Balance Sheet Snapshot

Automated exception alerts highlighting negative cash positions, equity erosion, and DSO policy breaches.

OFFICEMITRA

Books

Vault

data\ssdv.sqlite

As of

2026/03/31

Use ABC year-end 31 Mar 2026

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CEO pack

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POSTED BOOKS

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Scenario baselineAs of 2026-03-31Working capital 8,13,50,342.89

Download full report (HTML)

Download Board pack (PDF)

BOARD PACK

Red flags

DSO 156 days exceeds 120-day policy.

Cash position is negative (-2,497,907.63).

Equity after close is negative (-9,483,251.57).

Balance Sheet & Working Capital Breakdown

Visual bars analyzing Asset vs. Liability vs. Equity distribution and inventory/creditor/debtor components.

Last-month profit -446,884.17 is below prior month -312,028.49.

Assets

153,630,489.27

Liabilities

163,113,740.84

Equity after close

-9,483,251.57

Top vendor share

8%

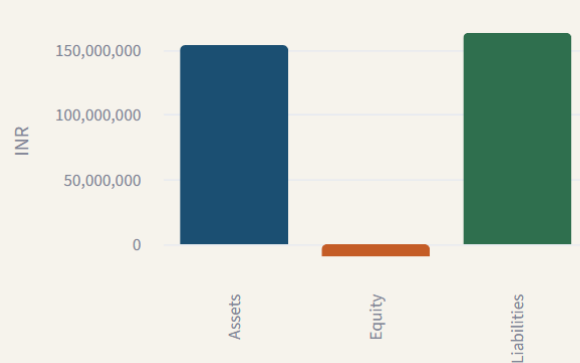
Gross margin

20.5%

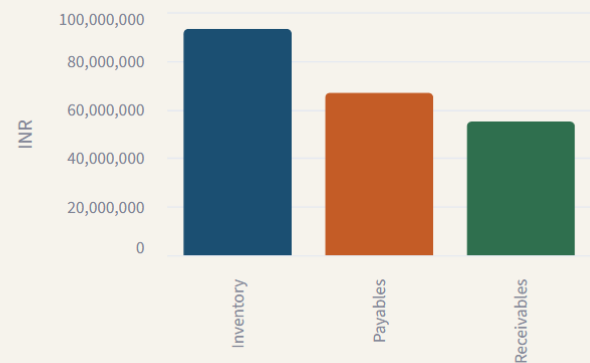
Cash conversion cycle

304 d

Balance sheet



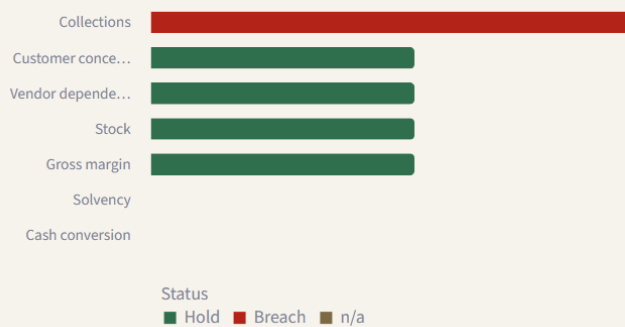
Working capital parts



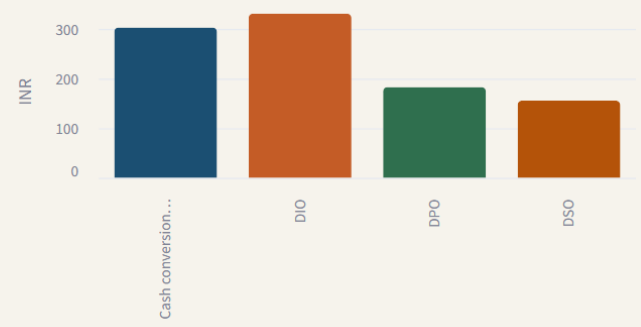
Enterprise Policy Scorecard

Traffic-light evaluation (Hold / Breach / Warning) across customer concentration, vendor dependency, collections, and solvency.

Policy scorecard



Cash conversion (days)



id	name	policy	actual	status	tone
customer_concentration	Customer concentration	Top customer < 35%	1%	Hold	success
vendor_dependency	Vendor dependency	Top vendor < 55%	8%	Hold	success
collections	Collections	DSO <= 120 days	156 d	Breach	danger
stock	Stock	Inventory / sales < 0.45	0.31	Hold	success
gross_margin	Gross margin	COGS / sales < 0.84	0.80	Hold	success
solvency	Solvency	Equity >= 0 after close	-9,483,251.57	Negative	danger
cash_stress	Cash conversion	Watch CCC and OD floor	304 d	OD drawn	warning

Benchmarks vs. Governance Standards

Formal compliance checklist comparing company performance against statutory and internal baseline standards.

Benchmarks					
SSDV policy bands for these trading books. Peer is ABC Industrial generated baseline (data/ssdv.sqlite) when that vault exists. Not a surveyed industry average.					
KPI	Actual	SSDV policy	Policy	ABC baseline	vs ABC
Gross margin	20.5%	at least 16.0%	Hold	n/a	No ABC baseline vault
COGS / sales	0.80	under 0.84	Hold	n/a	No ABC baseline vault
DSO	156 d	under 120 days	Breach	n/a	No ABC baseline vault
Collection efficiency	102.0%	at least 80.0%	Hold	n/a	No ABC baseline vault
Current ratio	0.97x	at least 1.00x	Breach	n/a	No ABC baseline vault
Quick ratio	0.33x	at least 1.00x	Breach	n/a	No ABC baseline vault
Inventory / sales	0.31	under 0.45	Hold	n/a	No ABC baseline vault
Top customer share	1.0%	under 35.0%	Hold	n/a	No ABC baseline vault
Cash conversion cycle	304 d	no SSDV day cap	n/a	n/a	No ABC baseline vault

Concentration Risk Disclosures (Customers & Vendors)

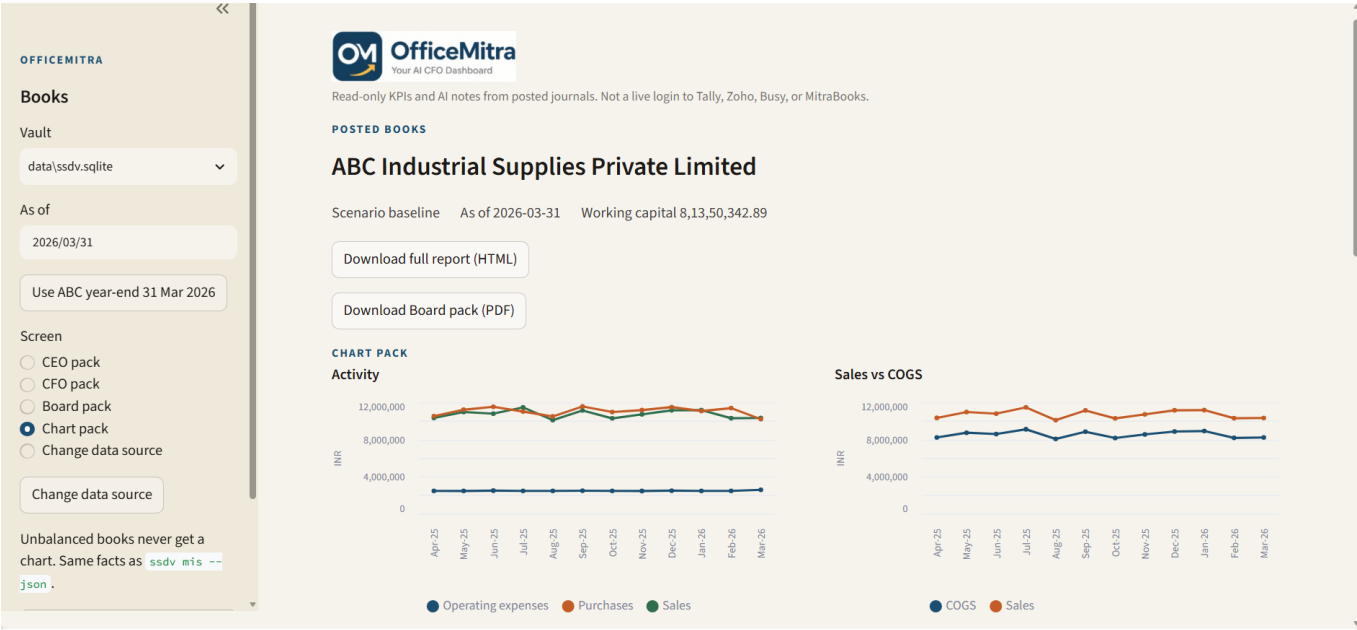
Tabular audit disclosures of top customer revenue shares and supplier dependencies.

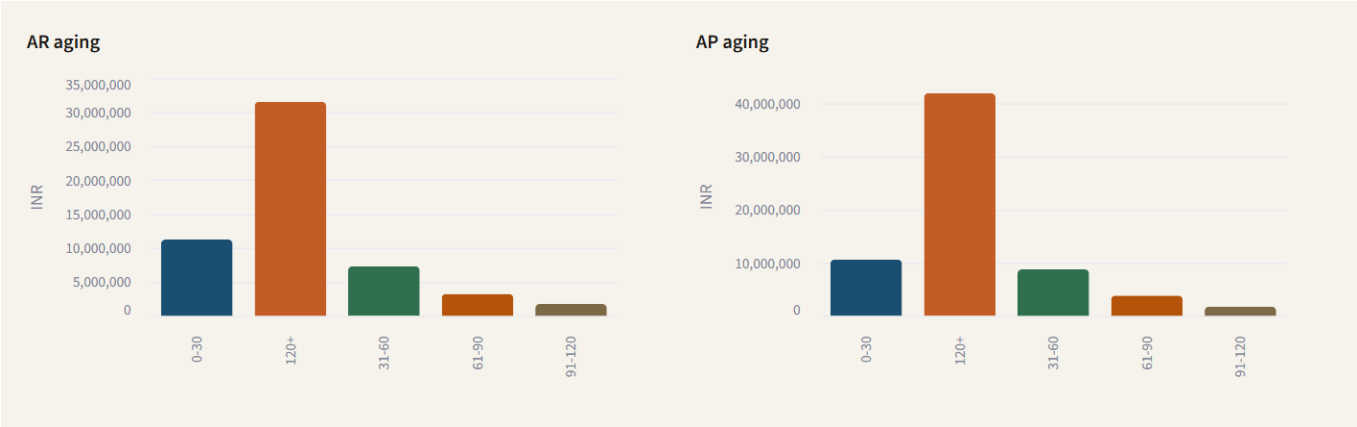
Customer & vendor intelligence					
Invoice aging 90+ where invoices exist. Journal parties with no invoice date are unaged. Customers ranked by AR 90+ (or largest AR if none). Vendors ranked by AP outstanding. Recommend-only; SSDV does not chase customers or pay vendors.					
Top overdue customers			Top vendor exposure		
Customer	AR 90+	Outstanding	Unaged	O	
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Vertex Switchgear Traders Pune	367007.82	656916.47	0.00		
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Summit Switchgear Traders Kolhapur	354670.19	439010.73	0.00		
Summit Power Solutions Solapur	310369.83	375098.48	0.00		
Vendor	Outstanding	AP 90+	Unaged	Oldes	
Precision Insulation Co	4318228.68	2594945.42	0.00		
Apex Polymers Pvt Ltd Kolhapur	4002062.15	2910379.18	0.00		
Continental Engineering Ltd	1713305.13	1103878.68	0.00		
Eastern Cable Mills Nashik	1682702.30	859797.43	0.00		
Universal Insulation Co Mumbai	1598378.70	1480308.42	0.00		

5.4 Chart Pack Dashboard

The **Chart Pack** visualizes 12-month operational trajectories:

Monthly Activity & Sales vs. COGS





5.5 Connect / Change Data Source Screen

The **Connect Screen** provides a simple, read-only data ingestion wizard:

Select ERP Data Source & Upload Journals

Upload standard CSV or XML daybooks from TallyPrime, Zoho Books, Busy, or generic accounting systems with custom ledger mapping.

OFFICEMITRA

Books

Vault

data/ssdv.sqlite

As of

2026/03/31

Use ABC year-end 31 Mar 2026

Screen

CEO pack

CFO pack

Board pack

Chart pack

Change data source

Change data source

Unbalanced books never get a chart. Same facts as `ssdv mis --json`.

Full-page screenshot

OM OfficeMitra

Your AI CFO Dashboard

Connect your accounts data

Choose where your books come from. OfficeMitra reads data only — it never writes back to Tally, Zoho, Busy, or any other application.

Data source

Any ERP (journal CSV) — Ready

Tally / TallyPrime — Ready

TallyPrime HTTP (live pull) — Ready

Zoho Books — Upload export file

Busy — Upload export file

Read-only. Export a day book / journal register to CSV (voucher_id, date, voucher_type, account, debit, credit) plus a ledger map.

Company name (shown on reports)

Journal / day-book CSV export

Upload200MB per file • CSV

Ledger map (optional CSV or YAML)

Upload200MB per file • CSV, YAML, YAML

Upload a file to continue.

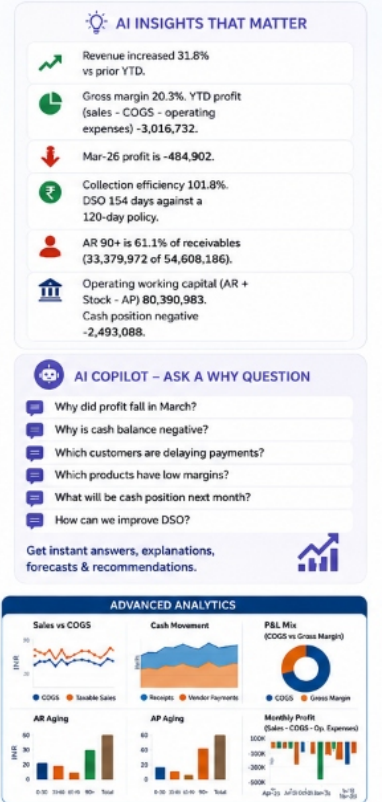
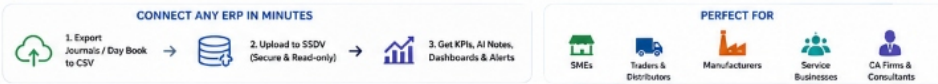
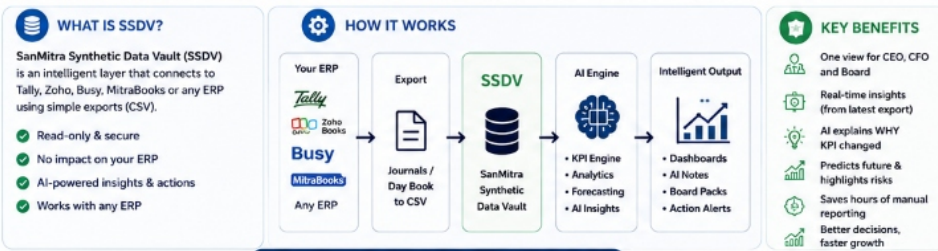
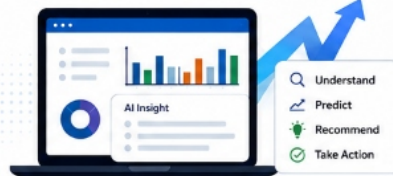
5.6 Executive MIS Architecture & AI Copilot Overview

OfficeMitra AI / SSDV

Your AI Business Analyst for SMEs

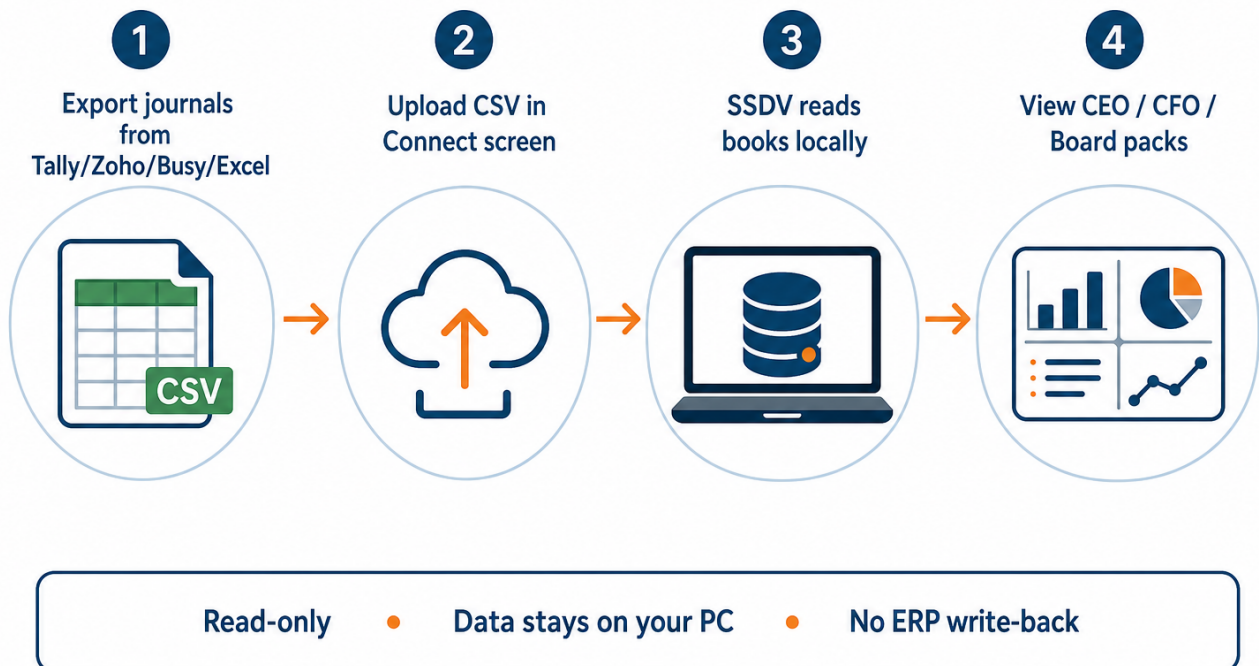
Read-only KPIs and AI insights from your ERP data.
Not a live login to Tally, Zoho, Busy, or MitraBooks.

Secure. Simple. Intelligent. Actionable.



6. Connect another application (read-only)

OfficeMitra — How it works



SSDV does **not** log into Tally / Zoho / Busy / MitraBooks. You **export** a journal / day book to CSV, map ledger names to SSDV account codes, then SSDV posts into a **sidecar** vault.

```
Any app --export CSV--> journals.csv + ledger_map.csv
--ssdv connect--> data\ssdv_connect.sqlite
--ssdv ui / mis--> KPIs, charts, AI notes
```

6.1 Which sources are ready

```
.\.venv\Scripts\ssdv.exe connect --list
```

id	Status today	What you do
`generic`	**ready**	Use this. CSV journals + map.
`tally`	export CSV	Export vouchers from Tally, then `--source generic`.
`zoho`	export CSV	Export journals from Zoho Books, then `--source generic`.
`busy`	export CSV	Export vouchers from Busy, then `--source generic`.
`mitrabooks`	export CSV	Export journals, or open an existing SSDV `.sqlite` in the Vault list.

6.2 Upload in the browser

1. Start the UI (any vault is fine).
2. Sidebar **Screen** → **Connect**.
3. **Journal CSV** — required. Shape: section 7.
4. **Ledger map** — recommended (CSV or YAML). Shape: section 7.
5. **Company name** — appears on packs (for example `Acme Trading Pvt Ltd`).
6. Tick **Replace books already in the connect vault** if `ssdv_connect.sqlite` already has data.
7. **Extract and post into SSDV**.
8. Sidebar **Screen** → **CEO**. Set **As of** to the last date in your file.

SSDV writes only to `data\ssdv_connect.sqlite`. It does not change `ssdv.sqlite`.

6.3 Connect from PowerShell

```
cd D:\SSDV
```



```

.\.venv\Scripts\ssdv.exe connect --source generic --journals C:\Exports\journals.csv --map C:\Exports\ledge
.\.venv\Scripts\ssdv.exe --db data/ssdv_connect.sqlite validate --as-of 2025-03-31
.\.venv\Scripts\ssdv.exe --db data/ssdv_connect.sqlite mis --as-of 2025-03-31 --pack ceo
.\.venv\Scripts\ssdv.exe --db data/ssdv_connect.sqlite ui --as-of 2025-03-31

```

Change `--as-of` to the last voucher date in **your** CSV.

`--force` wipes the connect vault and reloads. Without `--force`, a vault that already has vouchers is refused.

`ssdv ingest` is the same pipeline but writes `data\ssdv_ingest.sqlite` (older name). Prefer `ssdv connect`.

6.4 Tally / Zoho / Busy / MitraBooks in practice

Until native adapters exist, export a **journal register / day book** with one **row per debit or credit line** (not one row per voucher).

Typical Tally path: Display → Account Books → Journal / Day Book → export CSV. Rename columns to the names in section 7, or keep aliases SSDV already accepts (`ledger`, `dr`, `cr`, `vch_date`, ...).

Then:

```

.\.venv\Scripts\ssdv.exe connect --source generic --journals .\tally_daybook.csv --map .\ledger_map.csv --n

```

`--source tally` (or `zoho` / `busy` / `mitrabooks`) only prints the export hint and exits. That is expected.

Imported books have **no planted SSDV cause**. OfficeMitra may only explain numbers that are in the posted ledger.

7. File formats

Sample files: `examples\generic_ingest\journals.csv` and `examples\generic_ingest\ledger_map.csv`.

7.1 Journal CSV (required)

One row = one debit **or** one credit line. Lines that share the same `voucher_id` must **balance**.

Preferred columns

`voucher_id`, `date`, `voucher_type`, `account`, `debit`, `credit`, `party_type`, `party`, `narration`

Column	Required	Notes
<code>`voucher_id`</code>	yes	Same id on every line of one voucher. Aliases: <code>`voucher`</code> , <code>`id`</code> , <code>`vch_id`</code> .
<code>`date`</code>	yes	<code>`YYYY-MM-DD`</code> . Aliases: <code>`voucher_date`</code> , <code>`vch_date`</code> .
<code>`voucher_type`</code>	yes	See table below. Aliases: <code>`type`</code> , <code>`vch_type`</code> .

`account`	yes	Ledger name **or** SSDV account code. Aliases: `ledger`, `ledger_name`, `account_code`.
`debit`	yes	Amount or 0. Aliases: `dr`, `debit_amount`. Commas allowed.
`credit`	yes	Amount or 0. Aliases: `cr`, `credit_amount`.
`party_type`	no	`customer` or `vendor` on AR/AP lines.
`party`	no	Party code/name. Aliases: `party_code`, `party_name`.
`narration`	no	Aliases: `voucher_narration`.

Voucher types (aliases in brackets)

OPENING, PURCHASE (purchases), SALE (sales), RECEIPT (receipts), PAYMENT (payments), EXPENSE (expenses), JOURNAL, CONTRA, CREDIT_NOTE, DEBIT_NOTE, DEPRECIATION, CLOSING, GST_PAYMENT (gst).

Example (two-line opening)

```

voucher_id,date,voucher_type,account,debit,credit,party_type,party,narration
OPEN-1,2024-04-01,OPENING,HDFC Bank,500000.00,0,,,Opening
OPEN-1,2024-04-01,OPENING,Capital Account,0,500000.00,,,Opening

```

Rejected if:

- a voucher does not balance
- a ledger name is not in the map and is not a known account code
- the file path does not exist

7.2 Ledger map (strongly recommended)

CSV columns: source,account_code

source is the name as it appears in the journal CSV. account_code is an SSDV code from the chart of accounts.

```

source,account_code
HDFC Bank,111000
Sundry Debtors,120000
Sales Account,410000
Rent,620000
Capital Account,310000

```

YAML is also accepted (ledger: code). If you omit the map, every account value must already be a postable SSDV code (for example 111000).

7.3 Chart of accounts (map targets)

Headers (100000 Assets, 200000 Liabilities, ...) are **not** postable.

Code	Name
110000	Cash
111000	HDFC Bank Current
112000	ICICI Bank Current
113000	HDFC Bank OD
120000	Sundry Debtors
130000	Inventory
141000 / 141100 / 141200	Input CGST / SGST / IGST
151000	TDS Receivable
152000	Prepaid Expenses
161000 / 161100	Furniture / Accum. dep. furniture
162000 / 162100	Computers / Accum. dep. computers
210000	Sundry Creditors
221000 / 221100 / 221200	Output CGST / SGST / IGST
222000	GST Payable
231000	TDS Payable
232000	Salary Payable
233000	Expenses Payable
241000	Term Loan
310000	Share Capital
320000	Reserves & Surplus
330000	Profit & Loss Account
410000	Sales

410100	Sales Returns
420000 / 421000 / 422000	Other income / Interest / Discount received
510000	Cost of Goods Sold
520000 / 521000	Freight inward / Purchase returns
610000	Salaries
620000	Rent
621000–631000	Other operating expenses (electricity, insurance, bad debts, ...)
640000 / 641000 / 642000	Bank charges / Interest term loan / Interest OD
650000	Depreciation
660000	GST late fee

Full list: `src/ssdv/data/coa.yaml`.

8. PowerShell CLI reference

Global flag (before the subcommand):

```
.\.env\Scripts\ssdv.exe --db PATH\file.sqlite <command> ...
```

If you omit `--db`, generator commands use `data\ssdv.sqlite`. `connect` with no `--db` uses `data\ssdv_connect.sqlite`. `ingest` with no `--db` uses `data\ssdv_ingest.sqlite`.

`--scenario NAME` on generator commands writes `data\ssdv_NAME.sqlite` and does not overwrite the baseline vault.

8.1 Generate ABC books

```
.\.env\Scripts\ssdv.exe init
.\.env\Scripts\ssdv.exe purchases
.\.env\Scripts\ssdv.exe sales
.\.env\Scripts\ssdv.exe settle
.\.env\Scripts\ssdv.exe books
```

`init` already runs purchases, sales, settlements, and period books. The extra commands are additive if you run them later.

Danger: `init --force` **drops all tables** in that database. Do not use `--force` on the baseline vault unless you intend to rebuild ABC from scratch.

8.2 Check the books

```

.\.venv\Scripts\ssdv.exe validate --as-of 2026-03-31
.\.venv\Scripts\ssdv.exe tb --as-of 2026-03-31
.\.venv\Scripts\ssdv.exe equation --as-of 2026-03-31
.\.venv\Scripts\ssdv.exe aging --as-of 2026-03-31
.\.venv\Scripts\ssdv.exe ledger 130000 --as-of 2026-03-31

```

8.3 KPIs and dashboard without the browser

```

.\.venv\Scripts\ssdv.exe mis --as-of 2026-03-31
.\.venv\Scripts\ssdv.exe mis --pack ceo --as-of 2026-03-31
.\.venv\Scripts\ssdv.exe mis --pack cfo --as-of 2026-03-31
.\.venv\Scripts\ssdv.exe mis --pack board --as-of 2026-03-31
.\.venv\Scripts\ssdv.exe mis --pack ceo --json --as-of 2026-03-31
.\.venv\Scripts\ssdv.exe dashboard --as-of 2026-03-31
.\.venv\Scripts\ssdv.exe dashboard --pdf --as-of 2026-03-31

```

dashboard writes data\mis.html. Open that file in a browser.

dashboard --pdf writes data\board-pack.pdf (Board pack from posted journals).

--json is the contract for any other UI: tiles, monthly series, aging, insights.

8.4 Scenarios (known-cause seeds for OfficeMitra)

These are **separate** databases. They do not change ssdv.sqlite.

```

.\.venv\Scripts\ssdv.exe scenarios
.\.venv\Scripts\ssdv.exe init --scenario customer_concentration
.\.venv\Scripts\ssdv.exe explain --scenario customer_concentration --as-of 2026-03-31
.\.venv\Scripts\ssdv.exe --db data/ssdv_customer_concentration.sqlite ui --as-of 2026-03-31

```

Scenario	Known cause
`baseline`	Healthy growth
`customer_concentration`	Top customer about 40% of revenue
`cash_flow_crisis`	Sales up, DSO up, cash down
`inventory_buildup`	Inventory up, sales flat
`vendor_dependency`	One vendor dominates purchases
`margin_erosion`	Purchase cost up, selling price lags

On **imported** books, ignore planted causes. Explain only the measured KPIs.

8.5 Ask (local Ollama)

Start the **Ollama** app first so it listens on `http://127.0.0.1:11434`. Default model: `llama3.1:8b`.

```

.\.venv\Scripts\ssdv.exe ask "why is AR over 90 days up" --as-of 2026-03-31
.\.venv\Scripts\ssdv.exe mis --pack ceo --ask --as-of 2026-03-31
.\.venv\Scripts\ssdv.exe dashboard --as-of 2026-03-31 --ask

```

The model may use only the MIS JSON (posted facts). It must not invent a cause that is not in the books.

In Streamlit, CEO why chips (why did profit fall?, why is cash negative?, why is inventory rising?, why is AR over 90 days up?, which customers drive concentration?) call the same harness. A typed question still works. The chips and notes use posted journals only; they do not invent a customer or a cause.

8.6 Tests

```

cd D:\SSDV
.\.venv\Scripts\python.exe -m pytest tests -q

```

9. Glossary (what you see on the screen)

Term	Meaning
COGS	Cost of goods sold — cost of stock that was sold.
Opex / operating expenses	Running costs: salaries, rent, electricity, bank charges, depreciation. Not COGS.
YTD profit (sales – COGS – operating expenses)	Trading profit for the year to the as-of date from those flows. Not “cash in the bank”.
DSO	Days sales outstanding. How long customers take to pay. **AR ÷ FY sales × days** . Policy: within 120 days.
DIO	Days inventory outstanding. How long stock sits. **Inventory ÷ FY COGS × days** .
DPO	Days payable outstanding. How long you take to pay vendors. **AP ÷ FY purchases × days** .
Cash conversion cycle	**DSO + DIO – DPO** . Days **your** cash is tied up. Shorter is better.
Current ratio	Current assets ÷ current liabilities (term loan excluded).
Quick ratio	(Cash + AR) ÷ current liabilities.
Last-month net cash	Last month receipts – last month payments. Negative is a cash drain, not an opex guess.
Cash forecast (30 / 60 / 90)	Starting cash plus AR 0-30 / 31-60 / 61-90 minus AP in the same buckets. GST net liability and salary payable leave at 30 days. AR 90+ and AR with no invoice aging are not timed. No future sales.

GST net liability	Output GST – input GST from the trial balance (falls back to GST payable if those ledgers are empty).
Benchmarks	Hold / Breach vs SSDV policy bands. Optional peer is **ABC Industrial generated baseline** , not a surveyed industry average.
What-if	Recommend-only levers from posted books (DSO to 120, sales +15% at same margin, collect AR 90+, collections to 80%). Journals are not changed.
Top overdue customers	Ranked by AR 90+ from invoice aging. If there is no 90+ aging (CSV journals without invoice dates), ranked by largest AR balance. Unaged means no invoice date.
Top vendor exposure	Ranked by open AP. AP 90+ uses invoice/bill aging when bills exist.
Collection efficiency	Receipts ÷ sales for the period.
Operating working capital	AR + inventory – AP.
As of	The date of the snapshot. For ABC full books use **31 Mar 2026** . A date <i>after</i> year-end (for example 30 Apr 2026) shows an empty new year (sales 0, “revenue decreased 100%”).

10. Files on disk

Path	Role
`data\ssdv.sqlite`	ABC Industrial generated vault
`data\ssdv_connect.sqlite`	Connected / uploaded journals
`data\ssdv_ingest.sqlite`	Older ingest sidecar
`data\ssdv_.sqlite`	Scenario vaults
`data\mis.html`	Static CEO HTML from `ssdv dashboard`
`data\board-pack.pdf`	Board pack PDF from `ssdv dashboard --pdf`
`examples\generic_ingest\`	Sample CSV + map
`.venv\`	Python environment (not in Git)

SQLite files are **gitignored**. They stay on your PC; they are not on GitHub.

11. Troubleshooting

What you see	What to do
`Streamlit is not installed`	You used global `ssdv`. Run `.\.venv\Scripts\ssdv.exe ui ...`
Browser does not open	Leave the CLI window running; open http://localhost:8501
`cannot import name 'kpi_points`	Old Streamlit process. Ctrl+C, start UI again with the venv exe
`as_of cannot be modified after the widget`	Fixed in current code; refresh or restart UI, then use the year-end button
Notes say revenue decreased 100% / profit 0	As-of is after ABC year-end. Use **31 Mar 2026**
Connect / sample numbers on CEO	Vault is `ssdv_connect.sqlite`. Switch to `ssdv.sqlite` for ABC
`Unmapped ledger`	Add that name to the ledger map
`Unbalanced`	Debits $\neq$ credits for that `voucher_id`
Connect vault already has vouchers	Add `--force` or tick **Replace books**
`generic` / `Vouchers` / `Next:` errors in PowerShell	You pasted program output. Ignore it; type only real commands
Unified-Next preflight pytest mismatch	Do not pip-install SSDV or Streamlit into global Python
Edge screenshot is only the window	Download HTML report → open file → Ctrl+P → Save as PDF

12. Safety

- SSDV ****never**** posts into Tally, Zoho, Busy, or MitraBooks.
- Do not run `ssdv init --force` on the baseline vault unless you intend to rebuild it.
- Keep SSDV's `.venv` and `D:\sanmitra\_unified-Next\.venv` separate.
- OfficeMitra answers only from posted facts. On imported books it must not claim an SSDV “planted” scenario.

13. Command cheat sheet

```
cd D:\SSDV

# ABC Industrial
.\.venv\Scripts\ssdv.exe init
.\.venv\Scripts\ssdv.exe --db data/ssdv.sqlite ui --as-of 2026-03-31

# Sample CSV company
.\.venv\Scripts\ssdv.exe connect --source generic --journals examples/generic_ingest/journals.csv --map example
.\.venv\Scripts\ssdv.exe --db data/ssdv_connect.sqlite ui --as-of 2024-04-30

# Your export
```



```
.\.venv\Scripts\ssdv.exe connect --source generic --journals C:\Exports\journals.csv --map C:\Exports\ledge
.\.venv\Scripts\ssdv.exe --db data/ssdv_connect.sqlite mis --pack ceo --as-of YYYY-MM-DD
```

Blueprint (locked company rules): [BLUEPRINT.md](#).